Criterion C – Processing and Evaluating

Scenario

A toy company wants to find out which type of toy ball bounces the highest so they can choose the best material for making sports balls.

A group of students dropped three different balls (Tennis Ball, Rubber Ball and Foam Ball) from the **same height (100 cm)**. Each ball was dropped **three times**, and the bounce height was measured in centimetres.

As a scientist, your job is to process the data, analyse the results and evaluate the investigation.

Strand i: Presenting Data

Table 1: Raw Data

Ball Type	Trial 1 (cm)	Trial 2 (cm)	Trial 3 (cm)	Average (cm)
Tennis Ball	78	80	79	
Rubber Ball	91	90	92	
Foam Ball	42	44	43	

Your Task

Calculate the average bounce height for each ball.

Draw a **bar graph** using the average values.

Remember to:

- Give the graph a title.
- Label both axes.
- Include units.
- Use a suitable scale.

Strand ii: Interpret Data and Outline Results

Interpret the data shown in your table and bar graph.

Describe the relationship between the **type of ball** and the **bounce height**.

Identify any patterns or trends and support your answer with evidence from your data.

Strand iii: Discuss the Validity of a Prediction

Prediction

> **If different types of balls are dropped from the same height, then the rubber ball will bounce the highest because rubber is more elastic than foam or tennis balls.**

Discuss whether this prediction was supported by the results.

Use evidence from your data to justify your answer.

Strand iv: Discuss the Validity of the Method

Discuss the validity of the investigation.

Consider:

- Was it a fair test?
- Were all variables controlled?

- Was the data reliable?
- Were there any sources of error?

Strand v: Describe Improvements or Extensions

a) Describe two improvements that would make the investigation more accurate or reliable.

b) Suggest one extension to this investigation.

For example:

- Test different drop heights.
- Test different surfaces.
- Test heavier balls.
- Test worn and new balls.

Success Criteria (Checklist)

I calculated the averages correctly.

I drew a neat bar graph.

My graph has a title, labelled axes, units and a suitable scale.

I used evidence from my data to explain my answers.

I explained whether the prediction was supported.

I evaluated the investigation fairly.

I suggested realistic improvements and one extension.